

Solving the Measurement Problem: A Comprehensive Model Based on Three Postulates

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Abstract

The measurement problem is one of the most notorious and difficult open questions in all of physics. Why is it that quantum states seem to change their appearance just because of viewing or lack thereof? In the following paper, a possible solution will be explained using only three simple postulates which seek to generalize the concept of quantum superposition beyond just the subatomic world, and in the process remove any hints of questionable anthropocentric bias from the quantum theory. At its finest, this reasoning is potentially at once the missing piece to quantum theory that Einstein longed for [1], and the confirmation of correctness that Bohr was after [2].

1 Introduction to the Aethus

The fundamentals of Quantum Bayesianism [3], or QBism for short, will be assumed to be accurate for the sake of this paper's argument. In fact, they will be incorporated into a fundamental postulate, the first of three to be proposed in these pages, which, if accurate, will serve as a complete and sufficient explanation for the measurement problem.

This first postulate, named *the postulate of subjective reality*, argues that since QBism proposes that quantum states are only in effect relative to the observer, then in fact, through a generalization, all states in the universe are only in existence relative to an observer. This may seem inherently solipsistic, but it is not at all under the right interpretation. Instead, it shares more similarities with the basic argument that the universe may only be seen from *a perspective*. When absent of a perspective, all of reality loses its meaning. This argument works in much the same way.

In order to avoid a detrimental anthropocentric bias, we will give a name to the arbiter of reality, perspective, or what have you, which is human-independent. Let this reality-perceiving entity be referred to as an "Aethus". By virtue of the simple fact that you, the reader, are currently conscious and observing reality, you have an Aethus, which itself is directly representative and integrated with such things. An Aethus is a vantage point to the universe, but one which is strictly connected to it in both directions, such that reality loses meaning in the absence of an Aethus, and also an Aethus loses meaning in the absence of a corresponding reality to attach it to. In other words, reality comes in pairs – a reality coupled with its Aethus.

Postulate 1 (The Postulate of Subjective Reality) *Anything may only be real relative to an Aethus.*

A key next step is to declare what an Aethus actually stands for regarding a person's experience of reality. This is that a person's Aethus represents the collective presentation of all known truths that are currently accessible to them. That is, an Aethus is synonymous with a person's knowledge. Specifically, however, it is the deepest degree of such things, such that even if a person interprets said truths incorrectly and believes to know something differing from the contents of their Aethus, and still the Aethus remains unaffected by such things, all the while remaining definite and truthful depending exactly on what is accessible for certain.

For these reasons, an Aethus is very much like a set, in that it may contain or disclude "elements", so to speak. The difference is merely in the fact that an Aethus is somewhat more concrete, and is physically attached to an organic being so that it may perform organic tasks. For a human, the classic five senses, among other things, add to that human's Aethus by virtue of teaching them this or that, and so, in effect, they gain a fundamental knowledge of the world around them. So, of course,

other animals may have Aethae as well, as they are just as capable of collecting and storing mental information. Note that this does indeed open the doors to fictitious or collective Aethae. So long as a defining pattern is identifiable, an Aethus may be attached to it.

Note that an Aethus is constant and unchanging, which naturally means that the same creature or person in a different time possesses a different Aethus, since, afterall, the timecode is known to be different, which itself is separate information, hence implying a separate Aethus.

The Aethus now being officially defined as the representation of an entity's truthful and accessible knowledge, the next step is to determine whether or not any given Aethus is possible to exist.

2 Aethic Superposition

This section is where the first major deviation from our classic expectations for reality will be made. And yes, these deviations apply to all levels of scale, meaning that the assumption that the "quantum world" is somehow inherently different from the "classical world" will be argued against. The following propositions may seem to be some form of bizzare and unnatural, but in effect they make quantum physics not only more tangible, but perhaps more easy and straightforward to understand, because now there may finally be personal context to its notions.

The first argument is that of quantum superposition. The idea now is that any single object or concept will be in superposition, that is relative to an Aethus, (by the postulate of subjective reality), strictly following wherever it cannot be explained by said Aethus. That is, any attribute of reality which is independent of an Aethus, which simply put, means it exists "outside" the scope of said Aethus, will occupy all possible states at once that it may occupy without influencing the Aethus. For example, if your back is turned long enough for a parking lot to be filled with cars, then assuming your friend has parked in the lot, their car occupies every single space at once relative to your Aethus at the time¹. It is, in effect, in superposition. This of course sounds like an absurd proposition, as arguably humans are directly wired to object to such a concept, but it is what would be implied by the framework of the Aethus.

A natural objection that a human might make in the face of such a hypothesis, is that one may simply invalidate the claim that one's friend has parked, at least partially, in all spaces at once, by checking, and sure enough, seeing the car parked at a single location by one sensory mechanism or another. But, naturally, the Aethus framework has an explanation: simply by checking that the car is in a specific location, you have birthed a new Aethus entirely from your own Aethus, by simply adding the car's location to the mix. That is, you added the property of "car parking space index" to your Aethus, pulling a precisely random element from the superposition in the process, (as is the axiom). By definition, this new Aethus is a separate Aethus because its contents are not identical when comparing back and forth between them, however it may be argued that it is a kind of "child" Aethus of the old one, meaning it is identical except for added attributes. Even though the time code is different between the two Aethae, we may argue that this new one is still a child of the old one if we imagine attaching that old time code to itself as we might do with spatial displacement, (as in, at this time I held these coordinates. Now we say, at this time I held this time. This does mean that time is now two separate concepts — the coordinate time and the Aethic time, but this will be further elaborated on later). Note that this process is not as simple as simply ceasing the superposition of one concept with respect to one's Aethus — rather it is more like compressing the superposition in one particular way by virtue of "saturating" the Aethus with some hint of further information — but information that is nonetheless never entirely binary and fully known. For example, perhaps you mistake a similar looking car for that of your friend's, leading you to falsely believe your friend's car being in that spot is within your Aethus. By simply believing your friend's car is in that spot, your Aethus has nonetheless contracted the superposition of your friend's car, (assuming you have good judgment), because the likelihood of a car thought to be that of your friend's being that of your friend's is higher than the likelihood of a randomly selected car from the lot being that of your friend's, at least supposedly. This is all a way of saying that the superposition was still altered in some way, as now the Bayesian stakes are not identical, however still an attribute has not, nor ever will be, fully

¹One remaining open question regarding this, note, is how far the extent of the human senses reach. That is, perhaps all the needed information to deduce the car's location is perfectly available in the environment, and more, but a human simply never thinks to notice it for whatever reason. (Note that in the situation of quantum uncertainty this process is far more apparent because of a legitimate proof that some information is inaccessible, which in the macroscopic world is more vague). However, for the sake of the example we will equate the lack of visibility for the car with the spot of parking being absent from your own Aethus.

lifted out of superposition in one way or another.

Perhaps strangely, even though the parking location of your friend's car is now known, and thus is, at least roughly, "out of superposition", still you must remember that before you checked, you were unaware of that car's location, so even now, with the car in plain sight, it is in superposition relative to your Aethus of five minutes ago, or anytime before you physically checked. That is simply what such an Aethus would imply, given the nature of superposition being directly correlated to that which is unknown and independent of knowledgeable accessibility. In effect, because your Aethus has changed, so too has reality for you, because as mentioned earlier, a reality is only ever rooted to an Aethus by this framework, and is not objective. Your consciousness jumped realities so to speak.

The simple rule of an Aethus regarding superposition is that any concept in superposition relative to an Aethus has no sway on the observation of said Aethus in the moment, or in other words, is statistically independent to it. This automatically allows us to mark any Aethus which does not follow this rule as invalid, or in other words, it will forever stay on paper, and will never exist in our physical universe. Furthermore, we may postulate the converse of this basic principle, arguing that if an item is in superposition with respect to an Aethus, then it must be statistically independent to it. Again, breaking this calls for the invalidity of an Aethus, so by taking the contrapositive of the conjunction of both rules, a second major postulate of the Aethus is attainable.

Postulate 2 (The Postulate of Superposition) *If an Aethus is valid, then any given property will be in superposition relative to it if and only if it is statistically independent to it.*

It may seem that such a concept is sudden and hard to trace to previous evidence, but truly it is just an extrapolation of the concepts already defined in Aethic superposition.

The last and final postulate, now, will complete the picture of Aethic reasoning, and enable the measurement problem's solution.

3 Aethic Validity

One immediate response to the postulate of superposition might be to use it as a tool for identifying invalid Aethae. For instance, by simply taking the contrapositive of the statement, something of the following form is reached: *If any given property of an Aethus is statistically independent to it but not in superposition relative to it, or in superposition relative to it but not statistically independent to it, then such an Aethus would be invalid.* What this essentially argues for is two tools of determining the invalidity of an Aethus: either the Aethus contains a property which is also in superposition, or the Aethus doesn't contain a property which is not in superposition. These are two different arguments, but both point to an invalid Aethus. In this section, a third argument to test the invalidity will be postulated, which note is logically additional to the postulates named so far.

Using the claim that an Aethus containing a property which is in superposition relative to it is invalid, we may immediately modify the law of noncontradiction in a physical sense. Specifically, so long as an item is not possessed by an Aethus, it may exist in all cooperating states at once, which, of course, implies contradiction between them at least some of the time. The key, then, is that contradiction is merely prohibited only in the case of properties within an Aethus, or perhaps within an Aethus and also in superposition. That is, for the car example, we cannot claim that the car is in parking space *A* relative to an Aethus and is in superposition between parking spaces *A* through *Z* relative to the same Aethus, because, for instance, that would mean it is also in space *B*, *C*, *D*, and so on, which all differ from the Aethus's certainty of the car being in *A*.

Beyond here, we may start to build the final major postulate of Aethic logic, which may seem more complicated because it harnesses two generations of Aethae instead of one, but is nonetheless quite simple at its heart.

The general concept is that for any central Aethus, such as, for the sake of example, your own Aethus at this moment, it follows that, logically, it should be impossible for you to necessarily wind up in an invalid Aethus, as such an Aethus is invalid to begin with. We may think to perhaps define this as follows: "If an Aethus is valid, then no child Aethus of it may contradict any of its properties." This is close to the final realization, but still not quite accurate. To see why, we might argue that in order to prevent future Aethic invalidity through a child Aethus of your own, the universe diverts you away from such an invalid child Aethus by adding a property to your Aethus to prevent it from existing. That is, if there is only one obscure property which triggers a contradictory child Aethus to your own, then simply because there are other options of child Aethae to avoid that signifies that all

hope was not yet lost at the point of your own Aethus, hence proving that it is not invalid in the first place.

The solution to describing an Aethus which is necessarily doomed, so to speak, is sightless more specific. That is, it would have to be an Aethus for which no amount of property accumulation, no matter how clever, can prevent a potential contradiction. Such a concept may read as follows, if we are to take advantage of the family-style Aethus terminology. Consider this the event horizon of sorts of Aethus invalidity: “For some central Aethus, if for every one of its child Aethae there exists a further child Aethus that contradicts any of its properties, then said central Aethus is invalid.”

The last note to make here is that this only covers the hopeless situation of invalidity. However, if there are still options of child Aethae which necessarily prevent contradiction with your original Aethus, then it simply follows that one of these savior Aethae must in fact take the form of your own. If there are many as defined by many properties, then as we can say for certain is that they are all in superposition for taking the form of your own Aethus until a further child Aethus picks one over the other. (Also note that if the superposition described consists of one and only one property, then clearly said property will simply be added to one’s own Aethus accordingly.) Let us now finalize this postulate through presenting the contrapositive of this general statement. Here is this framework written out in the form of our final postulate for Aethic logic.

Postulate 3 (The Postulate of Aethic Certainty) *If some Aethus is valid, then there exists a child Aethus to it for which every one of its own child Aethae strictly do not contradict any of the properties of said Aethus.*

This general concept is logically sufficient to describe the remainder of this paper, however one interesting consequence of it may be quickly developed which implies that it has the potential to literally add to the Aethus itself. That is, given some central Aethus, if we are to remove all child Aethae of it which are invalid for their very attribute of being physically impossible, we are left with, oftentimes, a condensed set of potential child Aethae from any given Aethes compared to what may be generated by blindly adding properties to generate child Aethae. In effect, this puts the nature of all such condensed-possibility Aethae in superposition with the central Aethus itself. In the form of a statement this reads that, for some central Aethus, it follows that out of the set of all child Aethae to it which hold no further child Aethae that contradict any of the properties of the central Aethus, the defining additional properties to each one of the child Aethae in said set beyond the central Aethus must be in a mutually exhaustive superposition of existence relative to the original Aethus. If this set is empty, meaning for all child Aethae of the central Aethus, there exists a further child Aethus to each one of them which contradicts a property of the central Aethus, then said central Aethus is invalid. (With this last statement just being a rephrasing of the contrapositive of the postulate of Aethic certainty).

One last note about the yields of this postulate regards the state of transitivity of invalid Aethae to child Aethae. If we assume that, for the sake of argument, some central Aethus in question is to be invalid as per the postulate of Aethic certainty, then it follows that for all child Aethae to that central Aethus, there exists a further child Aethus which is invalid. This naturally implies that for each one of these “first generational” child Aethae to the original central Aethus, any one of their own child Aethae are also direct child Aethae to the central Aethus, simply due to the act of wrapping up both generations of property additions into a single property, which is a fully allowed. Supposing that an Aethus may be written as its own child if we are to add a kind of empty property, then it follows that for each first generational set of child Aethae to the central Aethus, they contain one, (being themselves), or more child Aethae of their own from that original first generational set. And, because it is guaranteed that every child Aethus from that first generational set has a child Aethus which contradicts a property of the central Aethus, and also that, by the definition of child Aethae, every Aethus in that first generational set also contains this same property, then it follows that all Aethae in the first generational set have all child Aethae which themselves must have contradictory child Aethueses of their own to the original property. This itself satisfies the condition of the postulate of Aethic certainty for an invalid Aethus, and so, in conclusion, every invalid Aethus must have all invalid child Aethae. To continue with the family vocabulary of Aethae, if we suppose that a “parent” Aethus is the inverse of a child, so an Aethus identical to a reference Aethus except with one or more removed properties, then it follows that your Aethus is invalid if any one of your parent Aethae is invalid.

4 Solution to the Measurement Problem

We are now ready to, with the help of a real-world thought experiment, solve the measurement problem itself as a tool to demonstrate the use of these three postulates. That real world example will be the double-slit experiment, so without further ado, let us continue onward.

To begin with, let us run the experiment in thought-experiment form. A beam of electrons is fired at a twice-pierced plate, which itself is in front of another plate. The experimenter and observer, (whose Aethus we will analyze), begins firing the electron beam, and sure enough, the back plate is lit up with an interference pattern. If we analyze the Aethus of the experimenter, then we may note that such an Aethus is unaware of which slit each electron is passing through, and so, by the logic supplied by the first two postulates, we may argue that such a beam is in superposition relative to that experimenter's Aethus. Therefore, each electron passes through both slits at once, and winds up creating the interference pattern by the logic of Feynman's path integral, (or in other words, the electrons pass through both slits, so in effect they are introducing half and half of their superposition into each slit, resulting in them collectively winding up in some spatial points more often than others). The back plate itself, we may argue, is like a kind of wave function collapse of sorts, because of the mere fact that where the electron touches this plate will be accessible to the Aethus of the experimenter. Therefore, as postulated before, when the experimenter updates their Aethus with the location of each electron on the back wall, they will randomly select an item from the superposition, compressing it accordingly. In this case, this means that the electron will select a random ending trajectory on the back plate as determined by its preexisting superposition of location there relative to the Aethus of the experimenter directly before they observed the resulting location.

Then, of course, comes the classic headache – when we boot on the detector and begin logging the slit that each next electron passes through. This is precisely where the Aethic framework is needed, because of the scenario where the experimenter destroys the data before they look at it.

Let us imagine that the experimenter, all while not looking at the pattern on the back plate, records the slit that each electron travels through on a detector, however the detector is nonetheless never played back to the eyes of the experimenter. As has been shown innumerable times, when the pattern on the back plate is checked, it will be seen that the interference pattern has disappeared, replaced by two regions of high concentration right behind the slits, exactly as would have happened had one analyzed every last electron's measurement from the detector instead of ignoring it. Why then, is an observation not even used counted by the universe as an observation nonetheless? What is the difference, truly, and a difference that is that forms the heart of the measurement problem. Funnily enough, with just the right manipulation, the solution becomes not only tangible, but intuitive due to Aethic reasoning.

The general idea here is to apply the postulate of Aethic certainty as a tool to observe what will happen. Let us suppose that you are the experimenter in our thought experiment, and you have run the detector during the double-slit experiment, which is now over. However, you have yet to check the pattern on the back plate, and are attempting to ponder, using Aethic reasoning as a tool, what you will observe. You start by subdividing the result into two potential Aethae: one in which there is an interference pattern, and another in which there is not. This corresponds to the notion, at least to an extremely high degree of likelihood, that the particle went through both slits at once, versus one slit but not the other in both ways, respectively speaking. For the sake of argument, you decide to begin by analyzing the validity of the Aethus where you see an interference pattern when you look. In such an Aethus, it follows that it contains many child Aethae in the universe, each one of which corresponds to a potential scenario for what you might do next, for example. However, to every one of those child Aethae, there is a potential for you to check the data on the detector, regardless of whether you actually do or not. That is, just due to the nature of quantum uncertainty, if a detector was not running, then it would be forever impossible to work out which slit the particles went through, but because it was running in this scenario, it is continuously possible to work out which single slit the particle went through. Therefore, according to the postulate of Aethic certainty, this hypothetical Aethus must be invalid, in effect adding to the Aethus that it cannot exist, or in other words, removing it from the superposition, (as per the postulate of superposition)². All that is now left, then, is for

²To further prove that the postulate of Aethic certainty is applicable here, we might suppose that the only means of creating a child Aethus of the experimenter's that does not contradict the observation of the interference pattern is to imagine a child Aethus in which the information of which slit the particles traveled through is permanently deleted from the universe, in which case it again becomes impossible to contradict the viewing of the interference pattern, hence supplying a child Aethus of the experimenter's with its own child Aethae as parallels the postulate of Aethic certainty. The problem with this, however, is that such a child Aethus holds a contradiction of its own to the original: it assumes

the particle to either go through one slit and not the other in either way. So long as the detector has not yet been checked, it follows that, relative to your Aethus, the particle went through one slit but not the other in both ways, as is the superposition. Your Aethus, then, passes the test of the postulate of Aethic certainty, because there exists child Aethae which have no further contradictory child Aethae of their own. Such child Aethae are the scenario where the particle went through, say, the “first” slit and not the second, and alternatively, through the second and not the first. If, for instance, it went through the first and not the second, then it follows that your own current Aethus is unaware of which specific slit it went through anyway, just knowing that it did not go through (at least) one of them. Hence, for either case of the particle going through the first slit or the second slit, as observed on the detector, no contradiction is raised, so either Aethus holds all non contradictory child Aethae, which disallows the postulate of Aethic certainty to make your Aethus invalid in this case. The only remaining possibility for invalidity, now, is through the postulate of superposition. But sure enough, it does not point to invalidity, because it is known that the particle may not go through both slits at once due to the previous work by the postulate of Aethic certainty on such a scenario, so the Aethus knows this – all that is left, then, is for the particle to go through the first slit and not the second slit as well as the second slit and not the first slit simultaneously in superposition. It may seem counterintuitive how this is different from the general going through both slits at once, but to Aethic logic it is certainly distinct – for instance, it does imply that one slit is not being passed through no matter what. Altogether the postulate of superposition processes this by triggering a superposition of the remaining unknowns, which themselves are the slit of traversal of the particle given that at least one slit was not traversed. Therefore, for such a child Aethus, no contradictions are possible through either postulate, and so the state of the particle going through one slit and not the other in both ways is valid, thus implying that this will be your observed result.

If, on the other hand, instead you are to keep the detector off during the running of the experiment, and then predict what pattern you shall see upon looking at the back plate, there are again several possibilities: either it goes through both slits at once, the first slit but not the second, the second but not the first, or neither. Because of the uncertainty principle, it follows that because no measurement was made of the particle’s slit of traversal, then that information is permanently lost, and will forever remain outside any child Aethus of your own. Therefore, due to this, every one of your child Aethae cannot contradict the slit index property of the particle within your own Aethus, and so the postulate of Aethic certainty has decidedly not proven your Aethus in that case to be invalid. All that is left, then, is to apply the contrapositive of the postulate of superposition as a final test to check the validity of your Aethus. If it also fails to prove your Aethus is invalid, then it must be valid, as it is physically impossible to prove it invalid. The only way, now, for your Aethus to be invalid, as according to this postulate, is for the particle slit to be known by your Aethus but also be in superposition, or be unknown by your Aethus and not be in superposition. Because we are fully aware that the particle’s slit is forever unknown in this case, it follows that, in order for your Aethus to be valid, it must be so that it is also in superposition. Therefore, the slit through which the particle passed is both slits at once, and hence the interference pattern is the direct result.

In effect, through the logic of the Aethus, the measurement problem is solved.

5 Applications

The measurement problem is solved by the three postulates of Aethic reasoning, however of course this does not prove that they themselves are correct.

Let us now demonstrate the power of Aethic reasoning by applying it to certain mysteries of quantum physics, starting with the noteworthy application of Aethic reasoning is its implications for quantum entanglement. Suppose that there are two people, Alice and Bob, who each possess an entangled particle held at opposite sides of the universe. If Alice checks her particle, then relative to her Aethus, Bob’s particle must possess the opposite state from wherever he is out in space. However, notably for the practical application of this, Bob’s particle, at least relative to his Aethus, will not collapse its superposition when Alice checks her particle, and likewise if Bob checks his particle, then that will not collapse the superposition of Alice’s particle then with respect to her Aethus. Moreover,

information may be deleted that clearly cannot be, as only the uncertainty principle would allow that, and in this case the uncertainty principle is clearly absent. Then, due to the definition of attribute inheritance in child Aethae, all child Aethae of this Aethus contradict the original, so the result is that a contradiction is utterly unavoidable for this particular experimenter’s Aethus, making it invalid through the postulate of Aethic certainty and implying that no interference pattern may be seen.

even after Alice has checked, relative to her Aethus from before she checked, both particles are still in an entangled superposition, and likewise, even after Bob has checked, relative to his Aethus from before, both particles are and always will be in an entangled state of superposition. And when checking the reverse case, we may also argue, likewise through a simple application of the postulate of subjective reality coupled with the postulate of superposition, that relative to Alice's current Aethus after the check, the particle was always in the state she now observes, as is the case with Bob's particle relative to his post-observational Aethus. Therefore, we are to get around the problem of hidden variables vs. non-local-realness [4] by simply stating that reality is what changed with the observation rather than any inherent property itself in a consistent reality.

A simpler version of this thought experiment is the notorious Schrödinger's cat itself, in which case a cat entangled to a particle is either alive or dead with respect to the Aethus of a human experimenter. In such a case, the experimenter, prior to observing the state of the cat, will not contain the wellbeing of the cat in their Aethus, and so therefore by the postulate of superposition such a thing will be in just that: superposition. This, hence, explains why the cat is in superposition. Moreover, once the experimenter checks the status of the cat, their Aethus will add the state from a random spot in the superposition, thus causing the outcome to update accordingly by whatever relevant probabilities are present. As stated many times before, relative to their prior Aethus the cat is still and always will be in superposition, and so whatever state they uncover with their new Aethus will merely be the state of the cat relative to this present Aethus, and will accordingly be so for the future and immediate past as well. If the cat is dead, then if the timecode of its death is ambiguous, perhaps such a thing may also be in superposition until determined, but for the meantime will be in superposition relative to the current Aethus.

One key concept to discuss beyond this is the Aethic implication for the nature of time. As was touched upon earlier, it is a requirement for the hypothesis of later versions of oneself holding child Aethae of previous versions that the time as a dimension be separated from the time perceived by one's Aethus. There are some interesting consequences to this notion, one of which is a solution to time asymmetry by the second law of thermodynamics, through stating that time symmetry was never broken in the first place. Specifically for this, the Aethic logic argues that the second law of thermodynamics is a property of the Aethus rather than of time. That is, when hypothesizing that each time one adds to their Aethus, such an added property is selected perfectly at random from the corresponding superposition, the second law of thermodynamics is easily derived from such a claim, just by arguing that heightening entropy is so much more likely than lowering entropy. A strange consequence of this, however, is that if somehow, hypothetically speaking, it was possible for time to reverse in an instant, then the universe would not merely play backwards, but entropy instead would continue to increase. That is, by the postulate of subjective reality, the very act of reversing the flow of time in the universe would only ever be a property of some Aethus, and so, relative to that Aethus, hypothetically the fundamental hypothesis that each new child Aethus randomly compresses the superposition would result in entropy continuing to increase relative to that Aethus. Therefore, the very act of reversing time would, in a sense, generate a new reality of time which has branched off of the old one, and decidedly would hold a time rate of negative magnitude relative to the original, a notion we may perhaps label as "Anti-Time", to reference an analogy to antimatter, but still would not simply backtrack the events of the old timeline in reverse. However, many quantities would still instantaneously reverse, such as velocity and magnetic field, while others would be unaffected, like position and acceleration in three dimensions. So, if, for example, a football was thrown and was midway through the air at the moment of time-reversal, the central observing Aethus, (your own, maybe), would observe the football's velocity vector directly flip, sending it, at least instantaneously, back the way it came. However, because of the maintaining of the second law of thermodynamics, air resistance would still curve its path away from a perfect parabola. There would be other strange effects in addition to all this as well. For instance, the day and year would start playing backwards, which effectively would amount to the same thing, but for the first few months would appear rather strange as summer turns to spring turns to winter. Eventually, Earth's biosphere would pick up on the change, and spring and fall would essentially reverse, but again the beginnings of it would be eerie. Not to mention that every textbook ever written about astronomy would describe an inverted plane of the solar system to the real one. There is also the question about what this would do to biology, as for a brief instant, all the blood in the human body would be flowing in the opposite direction. The human heart is not designed to pump in both ways, so it would resist this and try to force the blood back the other way, but for that instant where it skips a beat, the results could be catastrophic. This is not to even mention any further problems that might arise, such as unknown future physics effects

analogous to General Relativity that might create a gravitational field in the face of such an extreme instantaneous shift. But it is more or less apparent that if this inversion of time's direction were to be made, then all of us would be made of antimatter relative to ourselves of the previous direction, because after all, antimatter may be defined as inverted matter in time due to CPT symmetry.

While on the topic of time and alternate realities, a quick point of interest is to regard the Aethic perception of the multiverse. Rather than now being dimensionally based, the logic of this paper suggests, perhaps unsurprisingly, something more organic in a sense. I refer to it as "The Aethic Tree". The Aethic Tree is essentially the net product of recursively tracing out all valid parent and child Aethae to one's own, and repeating this process on each one of them infinitely. After this has been done, the Aethic Tree will contain every possible Aethus in the universe, in effect equating to every possible reality in the universe. I have now written multiple papers on the subject of what happens when one applies the mediocrity principle to one's location on the Aethic Tree, and the results are nothing short of fantastic. All such results, in fact, point to the Rare Earth Hypothesis³ [6], rather than the Copernican Principle [7] as does applying the mediocrity principle to our arbitrary location in space. The application of the mediocrity principle to the Aethic Tree is what I call "the accordance principle".

It is key in understanding the Aethic framework that it is interpreted in just the right way so as to not imply unrelated or contradictory conclusions. Namely, it argues for a quantum universe of utmost absence of anthropocentric bias rather than utmost abundance of it. Something of a paradox applies to the Copenhagen interpretation of quantum physics and the like which forces the logical deducer to make one anthropocentric claim or another, which note arises simply in the subtle disconnect between the observational phenomena and the human-based assumptions we always seem to make. In the case of Schrödinger's cat, for instance, the very nature of assuming that any human experimenter may collapse the wave function while the cat is but a metaphorical leaf in the wind places a heavy divide between the consciousness of the cat and the consciousness of a human. If this kind of reality were accurate, it would cause massive ripple effects in our nature of reality, where a human might as well be some kind of elementary entity in the universe. The point of this argument is to note not the nature of the human in this case, but instead to note the emphasis on a human versus any other creature such as a cat or alien. It is Cartesian and it lacks any semblance of natural compliance. This inherent ugliness implied by the Copenhagen interpretation is arguably what Einstein was referring to when he said that "god does not play dice with the universe" [8], or at least that is my interpretation of his words. While the Aethic framework may be argued by some to suffer the same problems, truly it is made to work as a kind of logical antithesis to this argument, and for such reasons I would argue that at its heart, it is chiefly oppositional to anthropocentrism. An Aethus now attaches to any organism or concept, to the point where the human aspect of things is no more fundamental than a pebble or a dragonfly. This is not to devalue humans, but instead to emphasize that they are but members of the cosmos along with everything else out there. The only place where humans truly become physically fundamental, then, is personal. You, the reader of this, in being a human, have such a thing linked to your Aethus, which serves as the center of your observed reality, and so relative to your reality, the humanity is fundamental, which is slightly different when viewed through the Aethus of a cat or a fish, etcetera, who would link their species to their Aethus instead. Furthermore, as seen with Bob and Alice, it would be highly anthropocentric by nature to suppose that either of them could collapse the superposition of an entangled pair of particles while a cat could not. It is as if every human is fundamentally linked up to the grand ticking clock of the universe, which is not only outwardly relativity defying, but again assumes a kind of Cartesian interpretation of the place of humans with respect to the rest of nature and the world. But with the Aethic perception, it is said that humans cause a wave function collapse not due to some ethereal status of ours, but instead because of the blatant state of only humans conferring with one another in all quantum studies, such that the human mind is a node in every last experiment we have or ever will do. Train a parrot to relay quantum information from experiments to humans, and sure enough, if the Aethic logic is correct, they too will collapse the wave function observationally like a human would. Generally speaking, the fact that all observations made by all observers are consistent is quite derivable from Aethic reasoning while not straying into a human-oriented universe perception. Back in the case of Bob and Alice, let us focus

³That's right! The Rare Earth Hypothesis! So, even assuming that the Aethic Tree exists, and is applicable with the mediocrity principle, automatically implies the Rare Earth Hypothesis – that is, given the proper data as well, which serves as the final aspect of my theorem for it, (something that we also have an abundance of). Therefore, the Aethic logic and a concrete solution to the Fermi Paradox [5] go hand in hand. That is, if the nature of the Aethus, as described in this paper is correct, then the Fermi Paradox is consequently solved, but otherwise it is not – at least for now.

on Alice’s Aethus for the sake of argument, but truly any substitute Aethus will suffice. Suppose that Alice has already checked her particle and found a corresponding spin, say A , thus implying that that is the spin of the particle relative to her Aethus. From here, we may either add to Alice’s Aethus that Bob sees B , the complement spin to A , or not. We may immediately derive that if he sees something other than B , no outside interference being incorporated, that such a version of Alice’s Aethus is invalid by the postulate of Aethic certainty, because it holds a contradiction within the Aethus, implying that so too will all its child Aethae. Therefore, all that is left as valid is for Bob to see B , and so such a thing is automatically added to Alice’s Aethus. In other words, their observations are consistent. This may be easily proven for any scenario using the same general logic of invalidating contradictory scenario Aethae. This serves as something of a further lemma that no two Aethae within/incorporated with a reality may hold opposing information to that present within the other, hence implying that all human experimenters are consistent when accurate, truth is unshakable, and an all-human quantum conference is easily mistakable for a sign of human physical fundamentality. Please also do note how this gives no definite logical implication to solipsism whatsoever, because firstly the mind is not involved with the Aethic system, and secondly an Aethus is perceived to attach to a reality full of Aethae, which is frankly not a stretch further at its core than the generally accepted statement of being an individual among the many of the world.

A further quick point of interest might be where human evolutionary buildup in certain regards has gradually diverted from the Aethic mindset fundamentally. One such example is the nature of imagining what is happening at some unseen location or time. By virtue of it being in superposition, at least in some way or another, compared to your current Aethus, it follows that actually observing the happenings of your pondering of choice would be physically impossible, as one can not, by definition, (postulate of superposition), observe a superposition. If anything, even the act of imagining what is happening at the location, such as the unseen parking lot from before, is in effect generating a pretend Aethus inside your own mind that may or may not be valid in practice. So even some of our fundamental intuitions of reality like the ability to imagine a subject, the linearity of time, or even a simple certainty that an object holds only one single state, become fundamentally questionable due to this underlying Aethic system of reality. It truly is incredible, and perhaps the ultimate victory of the strangeness of the quantum world that we are so quick to distance ourselves from. But, it seems, such strangeness is simply how reality functions on every scale.

One last note is that, as should hopefully be apparent by now, it is implied by the postulate of subjective reality that the Aethus itself is the basic unit of probabilistic reality rather than the workings of additively combining subatomic particles. As for the question of whether or not one is composed of the other or not, it should be assumed that the Aethus is more elemental, so to speak, (as would be implied by the postulate of subjective reality), until further research can provide more specificity. The reason is that proving otherwise may very well be an impossible task, on par to disproving solipsism or last Thursdayism [9], (for the sake of analogy). Also, as for how near the Aethus tends to Cartesian Dualism [10], it should be immediately noted that the Aethus is not quite synonymous with the mind, and if anything is more closely related to the fabric of reality than one’s feeling of it. However, eerily, the two are tied fundamentally at the point of one’s own knowledge intake, as is the work of the postulate of subjective reality. This paper argues that such a system is strictly implied by quantum physics, so the question is less to disprove it, (assuming it’s right at all), and more to understand what exactly is going on. It could be as simple as some flat correlation between observing and existing in reality, which is as simple as, “one’s observations are one’s observations”, and is otherwise no more complicated than some unifying one-level statement like that.

6 Conclusion

Only time will tell whether or not this Aethic logic is a necessary adaptation to quantum physics, or if it is a step in the wrong direction. But if any of this is relevant to the scene of the quantum universe, then truly the makeup of our reality is perhaps more strange than we had ever imagined. And in the meantime, hopefully some of the inherent ugliness of quantum physics has been directly addressed and solved with this systematic descriptive framework. Perhaps, for the first time, we can validate that “god does not play dice with the universe” [8].

Beyond this, perhaps further queries might build upon the foundation of this framework, such as the interaction between relativity and the Aethic mindset⁴, the meaning behind the two-dimensionality

⁴For now, my fundamental intuition is that there might be something important about the speed of light as pertains

of probability in quantum physics, and so on. I should also note that I am currently keeping the query of quantum gravity completely out of my headspace at the moment in case it does not lead me in the right instantaneous direction, however I have no objections to others working on this of course.

This all being said, thank you to those who have read this, and keep thinking.

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to superposition across space and time, however at the moment such ideas are heavily underdeveloped. I also have a feeling that special relativity and quantum physics may be able to be compiled at an even more basal level of fundamentality than what is currently done with quantum field theory, to the point where something like time as exists in current theories, for example, becomes simply an observational mirage of something deeper. Therefore linking two interpretations of time yields a messier result than linking one single pair-interpretation of the deeper concept in two ways. I feel that quantum physics and special relativity are one or more levels of fundamentality away from this basal stage. Obviously this is just brute intuition at this point, so actual results need to follow to demonstrate anything.